Fill in the table with the correct answer to the following (5) questions (10 - marks)

Q1) Let the $f(x)=\frac{\sqrt{x+5}-2}{\sqrt{1-x}}\frac{}{x}$. The natural domain is $a)[-5,1]\setminus\{0\}$ $b)[-5,\infty)\{0\}$ $c)[-5,1]\{0\}$ $d)(-\infty,6]$

Q2) Let $f(x)=2x^3-12x$ range f is $a)(-\infty,\infty)$ $b)(-\infty,-18)$ $c)[-18,\infty)$ $d)[-1,\infty)$

Q3) Let the $f(x)=\sqrt{x}$, $g(x)=x+1$ the natural domain of $(f\circ g)(x)$ is

$a)(-\infty,\infty)$ $b)(-\infty,-1)$ $c)(-\infty,6]$ $d)[-1,\infty)$

Q4) Let $f(x)=2x+x^3$ *find the value of* x *such that* $f^{-1}(x)=1$ a) 5 b) -5 c) 7

Q5) The $\cos^{-1}\left(\cos\frac{8\pi}{5}\right)$ is $a)2\pi/5$ $b)8\pi/5$ $c)8\pi$ $d)2\pi$

Essay Questions (20 marks)

Q6) find the natural domain for the function $f(x)=x^2+2\csc x$

Q7) If the function f is even and g is odd, show that $(fg)(x)$ is an odd function.

Q8) Let $f(x)=\sqrt[3]{3x-6}$ Find the formula of $f^{-1}(x)$

Q9) Let the graph of the function $f(x)$ is known, describe the graph of the function $g(x)=f(2-x)-4$

Q10) If the function $f(x)$ is shifted 5 units to the left and reflected about the x-axis, and then shift up 8 units, find the new function in terms of f.

- Best Wishes -

(5) $\cos^{-1}\left(\cos\frac{8\pi}{5}\right)$ $\qquad \cos x = \cos(2\pi - x)$

$\downarrow$

$2\pi - \frac{8\pi}{5}$

$\cos^{-1}\left(\cos\left(\frac{2\pi}{5}\right)\right) = \frac{2\pi}{5}$ (a)

نموذج (ب)

(6) $f(x) = x^2 + 2\csc x = x^2 + \frac{2}{\sin x}$ (4)

$D_f = \mathbb{R} \setminus \{\pm n\pi\}$, $n = 0, 1, 2, 3, \ldots$

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(7) f even $\Rightarrow f(-x) = f(x)$

g odd $\Rightarrow g(-x) = -g(x)$ (4)

$(fg)(x) \stackrel{?}{=} -fg(x)$

$f(-x)\, g(-x) = f(x)\,(-g(x)) = -(f\,g)(x)$ #

even odd

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(8) $y = \sqrt[3]{3x-6} \Rightarrow y^3 = 3x-6 \Rightarrow x = \frac{y^3+6}{3}$ (4)

$F^{-1}(x) = \frac{x^3+6}{3}$

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(9) $g(x) = +f(2-x) - 4$ (4)

$F(2-x)$ reflect about y-axis

Then $f(2-x)$ shift to the left 2 unit.

~~$-f(2-x)$ reflect about x-axis~~

$\#\, f(2-x) - 4$ shift down 4 unit.

(10) (4)

$f(x+5)$

$-f(x+5)$

$-f(x+5) + 8$